

# Technical Document #168: Machine Learning - Comprehensive Overview

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Neural networks are computing systems inspired by biological neural networks. They consist of layers of interconnected nodes that process information using connectionist approaches.

Random forests are ensemble learning methods that construct multiple decision trees during training and output the class that is the mode of the classes of individual trees.

Dimensionality reduction techniques like PCA and t-SNE help visualize high-dimensional data and can improve model performance by removing irrelevant features.

Support vector machines (SVMs) are supervised learning models that analyze data for classification and regression analysis. They find the hyperplane that best separates different classes.

Gradient descent is an optimization algorithm used to minimize the cost function in machine learning models. It iteratively adjusts parameters in the direction of steepest descent.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

### Key Takeaways:

- Random forests are ensemble learning methods that construct multiple decision trees during training and output the class that is the mode of the classes of individual trees.
- Ensemble methods combine multiple machine learning models to create a more powerful model.
- Neural networks are computing systems inspired by biological neural networks.